

pandas-4

September 29, 2025

```
[1]: import numpy as np
import pandas as pd
```

```
[2]: movies = pd.read_csv('movies.csv', index_col=0)
directors = pd.read_csv('directors.csv', index_col=0)
```

```
[ ]:
```

```
[3]: data = pd.merge(movies, directors, left_on='director_id', right_on='id',
↳ how='left')
```

```
[8]: data.head()
```

```
[8]:      budget  popularity    revenue \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
4  258000000         115   890871626

      title  vote_average  vote_count  year \
0      Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End       6.9        4500  2007
2      Spectre           6.3       4466  2015
3  The Dark Knight Rises           7.6       9106  2012
4  Spider-Man 3           5.9       3576  2007

      month    day  director_name  gender
0     Dec  Thursday    James Cameron   Male
1     May  Saturday    Gore Verbinski   Male
2     Oct   Monday      Sam Mendes   Male
3     Jul   Monday  Christopher Nolan   Male
4     May  Tuesday      Sam Raimi   Male
```

```
[7]: data.drop(columns=['id_x', 'id_y', 'director_id'], inplace=True)
```

```
[ ]:
```

```
[9]: # Get data of directors with risky movies
```

```
[10]: # Get data of movies with high budget directors.  
# If a movie has higher budget than the avg revenue by that director -> risky
```

```
[11]: def risky(g):  
# If a movies have budget > avg(revenue) of that director  
g['risky'] = g['budget'] - g['revenue'].mean() >= 0  
return g
```

```
[15]: 20 >= 50
```

```
[15]: False
```

```
[13]: data = data.groupby('director_name').apply(risky)
```

```
[14]: data.head()
```

```
[14]:      budget  popularity    revenue \  
0  237000000         150  2787965087  
1  300000000         139   961000000  
2  245000000         107   880674609  
3  250000000         112  1084939099  
4  258000000         115   890871626
```

```
      title  vote_average  vote_count  year \  
0      Avatar           7.2       11800  2009  
1  Pirates of the Caribbean: At World's End       6.9        4500  2007  
2      Spectre           6.3       4466  2015  
3  The Dark Knight Rises           7.6       9106  2012  
4  Spider-Man 3           5.9       3576  2007
```

```
      month    day  director_name  gender  risky  
0     Dec  Thursday    James Cameron   Male  False  
1     May  Saturday    Gore Verbinski   Male  False  
2     Oct   Monday      Sam Mendes    Male  False  
3     Jul   Monday  Christopher Nolan   Male  False  
4     May   Tuesday      Sam Raimi     Male  False
```

```
[17]: data['risky'].value_counts()
```

```
[17]: False    1334  
      True     131  
      Name: risky, dtype: int64
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

0.0.1 Most Productive Director

– More work more productive more movies

```
[18]: data.head()
```

```
[18]:
```

	budget	popularity	revenue	\
0	237000000	150	2787965087	
1	300000000	139	961000000	
2	245000000	107	880674609	
3	250000000	112	1084939099	
4	258000000	115	890871626	

	title	vote_average	vote_count	year	\
0	Avatar	7.2	11800	2009	
1	Pirates of the Caribbean: At World's End	6.9	4500	2007	
2	Spectre	6.3	4466	2015	
3	The Dark Knight Rises	7.6	9106	2012	
4	Spider-Man 3	5.9	3576	2007	

	month	day	director_name	gender	risky
0	Dec	Thursday	James Cameron	Male	False
1	May	Saturday	Gore Verbinski	Male	False
2	Oct	Monday	Sam Mendes	Male	False
3	Jul	Monday	Christopher Nolan	Male	False
4	May	Tuesday	Sam Raimi	Male	False

```
[19]: data['director_name'].value_counts()
```

```
[19]:
```

Steven Spielberg	26
Martin Scorsese	19
Clint Eastwood	19
Woody Allen	18
Ridley Scott	16
..	
Tim Hill	5
Jonathan Liebesman	5
Roman Polanski	5
Larry Charles	5
Nicole Holofcener	5

Name: director_name, Length: 199, dtype: int64

```
[ ]:
```

```
[20]: data.groupby('director_name').agg({'year' : ['min', 'max']})
```

```
[20]:
```

	year	
	min	max
director_name		
Adam McKay	2004	2015
Adam Shankman	2001	2012
Alejandro González Iñárritu	2000	2015
Alex Proyas	1994	2016
Alexander Payne	1999	2013
...	...	...
Wes Craven	1984	2011
Wolfgang Petersen	1981	2006
Woody Allen	1977	2013
Zack Snyder	2004	2016
Zhang Yimou	2002	2014

[199 rows x 2 columns]

```
[22]: data_agg = data.groupby('director_name').agg({'year' : ['min', 'max'], 'title' :  
↪ 'count'})
```

```
[23]: data_agg
```

```
[23]:
```

	year		title
	min	max	count
director_name			
Adam McKay	2004	2015	6
Adam Shankman	2001	2012	8
Alejandro González Iñárritu	2000	2015	6
Alex Proyas	1994	2016	5
Alexander Payne	1999	2013	5
...	...	...	...
Wes Craven	1984	2011	10
Wolfgang Petersen	1981	2006	7
Woody Allen	1977	2013	18
Zack Snyder	2004	2016	7
Zhang Yimou	2002	2014	6

[199 rows x 3 columns]

```
[24]: data_agg.columns
```

```
[24]: MultiIndex([( 'year',   'min'),  
                ( 'year',   'max'),
```

```

        ('title', 'count']],
    )

```

```
[25]: data_agg['year']
```

```
[25]:
```

	min	max
director_name		
Adam McKay	2004	2015
Adam Shankman	2001	2012
Alejandro González Iñárritu	2000	2015
Alex Proyas	1994	2016
Alexander Payne	1999	2013
...	...	...
Wes Craven	1984	2011
Wolfgang Petersen	1981	2006
Woody Allen	1977	2013
Zack Snyder	2004	2016
Zhang Yimou	2002	2014

[199 rows x 2 columns]

```
[27]: data_agg['year']['max']
```

```
[27]:
```

director_name	
Adam McKay	2015
Adam Shankman	2012
Alejandro González Iñárritu	2015
Alex Proyas	2016
Alexander Payne	2013
...	...
Wes Craven	2011
Wolfgang Petersen	2006
Woody Allen	2013
Zack Snyder	2016
Zhang Yimou	2014

Name: max, Length: 199, dtype: int64

```
[30]: data_agg['title']['count']
```

```
[30]:
```

director_name	
Adam McKay	6
Adam Shankman	8
Alejandro González Iñárritu	6
Alex Proyas	5
Alexander Payne	5
...	..
Wes Craven	10

```

Wolfgang Petersen      7
Woody Allen            18
Zack Snyder            7
Zhang Yimou            6
Name: count, Length: 199, dtype: int64

```

```
[31]: data_agg[('title', 'count')]
```

```

[31]: director_name
Adam McKay            6
Adam Shankman         8
Alejandro González Iñárritu 6
Alex Proyas           5
Alexander Payne       5
..
Wes Craven           10
Wolfgang Petersen    7
Woody Allen          18
Zack Snyder          7
Zhang Yimou          6
Name: (title, count), Length: 199, dtype: int64

```

```
[44]: data_agg.head()
```

```

[44]:      director_name  year_min  year_max  title_count
0      Adam McKay      2004      2015         6
1      Adam Shankman    2001      2012         8
2  Alejandro González Iñárritu  2000      2015         6
3      Alex Proyas      1994      2016         5
4      Alexander Payne  1999      2013         5

```

```
[33]: # Converting the columns in single level indexing
```

```
[34]: data_agg.columns
```

```

[34]: MultiIndex([( 'year',  'min'),
                  ( 'year',  'max'),
                  ('title', 'count')],
                )

```

```
[35]: data_agg.columns = ['year_min', 'year_max', 'title_count']
```

```
[45]: data_agg.head()
```

```

[45]:      director_name  year_min  year_max  title_count
0      Adam McKay      2004      2015         6
1      Adam Shankman    2001      2012         8

```

2	Alejandro González Iñárritu	2000	2015	6
3	Alex Proyas	1994	2016	5
4	Alexander Payne	1999	2013	5

```
[46]: data_agg.reset_index(inplace=True)
```

```
[48]: data_agg.head()
```

```
[48]:
```

	index	director_name	year_min	year_max	title_count
0	0	Adam McKay	2004	2015	6
1	1	Adam Shankman	2001	2012	8
2	2	Alejandro González Iñárritu	2000	2015	6
3	3	Alex Proyas	1994	2016	5
4	4	Alexander Payne	1999	2013	5

```
[49]: data_agg['active_year'] = data_agg['year_max'] - data_agg['year_min']
```

```
[50]: data_agg.head()
```

```
[50]:
```

	index	director_name	year_min	year_max	title_count	\
0	0	Adam McKay	2004	2015	6	
1	1	Adam Shankman	2001	2012	8	
2	2	Alejandro González Iñárritu	2000	2015	6	
3	3	Alex Proyas	1994	2016	5	
4	4	Alexander Payne	1999	2013	5	

	active_year
0	11
1	11
2	15
3	22
4	14

```
[51]: data_agg['movies_per_year'] = data_agg['title_count']/data_agg['active_year']
```

```
[52]: data_agg.head()
```

```
[52]:
```

	index	director_name	year_min	year_max	title_count	\
0	0	Adam McKay	2004	2015	6	
1	1	Adam Shankman	2001	2012	8	
2	2	Alejandro González Iñárritu	2000	2015	6	
3	3	Alex Proyas	1994	2016	5	
4	4	Alexander Payne	1999	2013	5	

	active_year	movies_per_year
0	11	0.545455
1	11	0.727273

2	15	0.400000
3	22	0.227273
4	14	0.357143

```
[60]: data_agg.sort_values(by = 'movies_per_year', ascending=False)
```

```
[60]:
```

	index	director_name	year_min	year_max	title_count	active_year	\
	190	Tyler Perry	2006	2013	9	7	
	73	Jason Friedberg	2006	2010	5	4	
	169	Shawn Levy	2002	2014	11	12	
	158	Robert Rodriguez	1992	2014	16	22	
	1	Adam Shankman	2001	2012	8	11	
..	...	...	...	...	...	...	
	104	Lawrence Kasdan	1985	2012	5	27	
	109	Luc Besson	1985	2014	5	29	
	157	Robert Redford	1980	2010	5	30	
	170	Sidney Lumet	1976	2006	5	30	
	117	Michael Apted	1980	2010	5	30	

	movies_per_year
190	1.285714
73	1.250000
169	0.916667
158	0.727273
1	0.727273
..	...
104	0.185185
109	0.172414
157	0.166667
170	0.166667
117	0.166667

[199 rows x 7 columns]

```
[62]: data_agg.sort_values(by = 'movies_per_year', ascending=False).
      ↪iloc[0]['director_name']
```

```
[62]: 'Tyler Perry'
```

```
[63]: # HW: Get only those directors in output who are in businees for atleast 10
      ↪years?
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```



```
[ ]:
```

```
[ ]:
```

0.1 Working with Pfizer dataset

- <https://drive.google.com/file/d/173A59xh2mnpmljCCB9bhC4C5eP2IS6qZ/view>

```
[78]: # Melt
```

```
[65]: data = pd.read_csv('Pfizer_1.csv')
```

```
[66]: data.head()
```

```
[66]:
```

	Date	Drug_Name	Parameter	1:30:00	2:30:00	\
0	15-10-2020	diltiazem hydrochloride	Temperature	23.0	22.0	
1	15-10-2020	diltiazem hydrochloride	Pressure	12.0	13.0	
2	15-10-2020	docetaxel injection	Temperature	NaN	17.0	
3	15-10-2020	docetaxel injection	Pressure	NaN	22.0	
4	15-10-2020	ketamine hydrochloride	Temperature	24.0	NaN	

	3:30:00	4:30:00	5:30:00	6:30:00	7:30:00	8:30:00	9:30:00	10:30:00	\
0	NaN	21.0	21.0	22	23.0	21.0	22.0	20	
1	NaN	11.0	13.0	14	16.0	16.0	24.0	18	
2	18.0	NaN	17.0	18	NaN	NaN	23.0	23	
3	22.0	NaN	22.0	23	NaN	NaN	27.0	26	
4	NaN	27.0	NaN	26	25.0	24.0	23.0	22	

	11:30:00	12:30:00
0	20.0	21
1	19.0	20
2	25.0	25
3	29.0	28
4	21.0	20

```
[68]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18 entries, 0 to 17
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Date        18 non-null    object
1   Drug_Name   18 non-null    object
2   Parameter   18 non-null    object
3   1:30:00     16 non-null    float64
4   2:30:00     16 non-null    float64
```

```

5   3:30:00    12 non-null    float64
6   4:30:00    14 non-null    float64
7   5:30:00    16 non-null    float64
8   6:30:00    18 non-null    int64
9   7:30:00    16 non-null    float64
10  8:30:00    14 non-null    float64
11  9:30:00    16 non-null    float64
12 10:30:00    18 non-null    int64
13 11:30:00    16 non-null    float64
14 12:30:00    18 non-null    int64
dtypes: float64(9), int64(3), object(3)
memory usage: 2.2+ KB

```

```
[69]: data.shape
```

```
[69]: (18, 15)
```

```
[71]: pd.melt(data, id_vars = ['Date', 'Drug_Name', 'Parameter'])
```

```

[71]:
      Date      Drug_Name  Parameter  variable  value
0  15-10-2020  diltiazem hydrochloride  Temperature    1:30:00    23.0
1  15-10-2020  diltiazem hydrochloride    Pressure    1:30:00    12.0
2  15-10-2020      docetaxel injection  Temperature    1:30:00     NaN
3  15-10-2020      docetaxel injection    Pressure    1:30:00     NaN
4  15-10-2020  ketamine hydrochloride  Temperature    1:30:00    24.0
..      ...
211 17-10-2020  diltiazem hydrochloride    Pressure   12:30:00    14.0
212 17-10-2020      docetaxel injection  Temperature   12:30:00    23.0
213 17-10-2020      docetaxel injection    Pressure   12:30:00    28.0
214 17-10-2020  ketamine hydrochloride  Temperature   12:30:00    24.0
215 17-10-2020  ketamine hydrochloride    Pressure   12:30:00    15.0

```

```
[216 rows x 5 columns]
```

```
[80]: data_melt = pd.melt(data, id_vars = ['Date', 'Drug_Name', 'Parameter'],
                        var_name='time', value_name='readings')
```

```
[81]: data_melt.head()
```

```

[81]:
      Date      Drug_Name  Parameter    time  readings
0  15-10-2020  diltiazem hydrochloride  Temperature  1:30:00    23.0
1  15-10-2020  diltiazem hydrochloride    Pressure  1:30:00    12.0
2  15-10-2020      docetaxel injection  Temperature  1:30:00     NaN
3  15-10-2020      docetaxel injection    Pressure  1:30:00     NaN
4  15-10-2020  ketamine hydrochloride  Temperature  1:30:00    24.0

```

```
[82]: data.head()
```

```
[82]:
```

	Date	Drug_Name	Parameter	1:30:00	2:30:00	\
0	15-10-2020	diltiazem hydrochloride	Temperature	23.0	22.0	
1	15-10-2020	diltiazem hydrochloride	Pressure	12.0	13.0	
2	15-10-2020	docetaxel injection	Temperature	NaN	17.0	
3	15-10-2020	docetaxel injection	Pressure	NaN	22.0	
4	15-10-2020	ketamine hydrochloride	Temperature	24.0	NaN	

	3:30:00	4:30:00	5:30:00	6:30:00	7:30:00	8:30:00	9:30:00	10:30:00	\
0	NaN	21.0	21.0	22	23.0	21.0	22.0	20	
1	NaN	11.0	13.0	14	16.0	16.0	24.0	18	
2	18.0	NaN	17.0	18	NaN	NaN	23.0	23	
3	22.0	NaN	22.0	23	NaN	NaN	27.0	26	
4	NaN	27.0	NaN	26	25.0	24.0	23.0	22	

	11:30:00	12:30:00
0	20.0	21
1	19.0	20
2	25.0	25
3	29.0	28
4	21.0	20

```
[ ]:
```

Pivoting

```
[83]: data_melt.head()
```

```
[83]:
```

	Date	Drug_Name	Parameter	time	readings
0	15-10-2020	diltiazem hydrochloride	Temperature	1:30:00	23.0
1	15-10-2020	diltiazem hydrochloride	Pressure	1:30:00	12.0
2	15-10-2020	docetaxel injection	Temperature	1:30:00	NaN
3	15-10-2020	docetaxel injection	Pressure	1:30:00	NaN
4	15-10-2020	ketamine hydrochloride	Temperature	1:30:00	24.0

```
[86]: data_melt.pivot(index=['Date', 'Drug_Name', 'Parameter'],
                        columns='time',
                        values='readings')
```

```
[86]:
```

time	Date	Drug_Name	Parameter	10:30:00	11:30:00	12:30:00	\
	15-10-2020	diltiazem hydrochloride	Pressure	18.0	19.0	20.0	
			Temperature	20.0	20.0	21.0	
		docetaxel injection	Pressure	26.0	29.0	28.0	
			Temperature	23.0	25.0	25.0	
		ketamine hydrochloride	Pressure	9.0	9.0	11.0	
			Temperature	22.0	21.0	20.0	
	16-10-2020	diltiazem hydrochloride	Pressure	24.0	NaN	27.0	

		Temperature	40.0	NaN	42.0
	docetaxel injection	Pressure	28.0	29.0	30.0
		Temperature	56.0	57.0	58.0
	ketamine hydrochloride	Pressure	16.0	17.0	18.0
		Temperature	13.0	14.0	15.0
17-10-2020	diltiazem hydrochloride	Pressure	11.0	13.0	14.0
		Temperature	14.0	11.0	10.0
	docetaxel injection	Pressure	28.0	29.0	28.0
		Temperature	21.0	22.0	23.0
	ketamine hydrochloride	Pressure	13.0	14.0	15.0
		Temperature	22.0	23.0	24.0

time 1:30:00 2:30:00 3:30:00 \

Date	Drug_Name	Parameter			
15-10-2020	diltiazem hydrochloride	Pressure	12.0	13.0	NaN
		Temperature	23.0	22.0	NaN
	docetaxel injection	Pressure	NaN	22.0	22.0
		Temperature	NaN	17.0	18.0
	ketamine hydrochloride	Pressure	8.0	NaN	NaN
		Temperature	24.0	NaN	NaN
16-10-2020	diltiazem hydrochloride	Pressure	18.0	19.0	20.0
		Temperature	34.0	35.0	36.0
	docetaxel injection	Pressure	23.0	24.0	NaN
		Temperature	46.0	47.0	NaN
	ketamine hydrochloride	Pressure	12.0	12.0	13.0
		Temperature	8.0	9.0	10.0
17-10-2020	diltiazem hydrochloride	Pressure	3.0	4.0	4.0
		Temperature	20.0	19.0	19.0
	docetaxel injection	Pressure	20.0	22.0	22.0
		Temperature	12.0	13.0	14.0
	ketamine hydrochloride	Pressure	8.0	9.0	10.0
		Temperature	13.0	14.0	15.0

time 4:30:00 5:30:00 6:30:00 \

Date	Drug_Name	Parameter			
15-10-2020	diltiazem hydrochloride	Pressure	11.0	13.0	14.0
		Temperature	21.0	21.0	22.0
	docetaxel injection	Pressure	NaN	22.0	23.0
		Temperature	NaN	17.0	18.0
	ketamine hydrochloride	Pressure	7.0	NaN	9.0
		Temperature	27.0	NaN	26.0
16-10-2020	diltiazem hydrochloride	Pressure	21.0	22.0	23.0
		Temperature	36.0	37.0	38.0
	docetaxel injection	Pressure	25.0	26.0	27.0
		Temperature	48.0	48.0	49.0
	ketamine hydrochloride	Pressure	NaN	15.0	15.0
		Temperature	NaN	11.0	12.0

17-10-2020	diltiazem hydrochloride	Pressure	4.0	6.0	8.0
		Temperature	18.0	17.0	16.0
	docetaxel injection	Pressure	22.0	22.0	23.0
		Temperature	15.0	16.0	17.0
	ketamine hydrochloride	Pressure	11.0	11.0	12.0
		Temperature	16.0	17.0	18.0
time			7:30:00	8:30:00	9:30:00
Date	Drug_Name	Parameter			
15-10-2020	diltiazem hydrochloride	Pressure	16.0	16.0	24.0
		Temperature	23.0	21.0	22.0
	docetaxel injection	Pressure	NaN	NaN	27.0
		Temperature	NaN	NaN	23.0
	ketamine hydrochloride	Pressure	10.0	11.0	10.0
		Temperature	25.0	24.0	23.0
16-10-2020	diltiazem hydrochloride	Pressure	24.0	25.0	25.0
		Temperature	37.0	38.0	39.0
	docetaxel injection	Pressure	28.0	29.0	28.0
		Temperature	50.0	52.0	55.0
	ketamine hydrochloride	Pressure	15.0	15.0	NaN
		Temperature	12.0	11.0	NaN
17-10-2020	diltiazem hydrochloride	Pressure	9.0	NaN	9.0
		Temperature	15.0	NaN	13.0
	docetaxel injection	Pressure	25.0	26.0	27.0
		Temperature	18.0	19.0	20.0
	ketamine hydrochloride	Pressure	12.0	11.0	12.0
		Temperature	19.0	20.0	21.0

```
[89]: tidy_data = data_melt.pivot(index=['Date', 'Drug_Name', 'time'],
                                columns = 'Parameter',
                                values = 'readings')
```

```
[90]: tidy_data.head()
```

```
[90]: Parameter
Date      Drug_Name      time      Pressure  Temperature
15-10-2020 diltiazem hydrochloride 10:30:00      18.0          20.0
                                     11:30:00      19.0          20.0
                                     12:30:00      20.0          21.0
                                     1:30:00       12.0          23.0
                                     2:30:00      13.0          22.0
```

```
[92]: tidy_data.reset_index(inplace=True)
```

```
[93]: tidy_data.head()
```

```
[93]: Parameter      Date      Drug_Name      time  Pressure  \
0      15-10-2020  diltiazem hydrochloride  10:30:00      18.0
1      15-10-2020  diltiazem hydrochloride  11:30:00      19.0
2      15-10-2020  diltiazem hydrochloride  12:30:00      20.0
3      15-10-2020  diltiazem hydrochloride   1:30:00      12.0
4      15-10-2020  diltiazem hydrochloride   2:30:00      13.0

Parameter  Temperature
0          20.0
1          20.0
2          21.0
3          23.0
4          22.0
```

```
[ ]:
```

Binning of data:

- Getting categories from numerical

```
[96]: tidy_data.head()
```

```
[96]: Parameter      Date      Drug_Name      time  Pressure  \
0      15-10-2020  diltiazem hydrochloride  10:30:00      18.0
1      15-10-2020  diltiazem hydrochloride  11:30:00      19.0
2      15-10-2020  diltiazem hydrochloride  12:30:00      20.0
3      15-10-2020  diltiazem hydrochloride   1:30:00      12.0
4      15-10-2020  diltiazem hydrochloride   2:30:00      13.0

Parameter  Temperature
0          20.0
1          20.0
2          21.0
3          23.0
4          22.0
```

```
[98]: # pd.cut()
```

```
[99]: tidy_data['Temperature'].min()
```

```
[99]: 8.0
```

```
[100]: tidy_data['Temperature'].max()
```

```
[100]: 58.0
```

```
[101]: temp_points = [5, 20, 35, 50, 60]
temp_labels = ['low', 'medium', 'high', 'very_high']

[109]: pd.cut(tidy_data['Temperature'], bins=[5, 20, 35, 50, 60], labels=['low', 'medium', 'high', 'very_high'])
```

```
[109]: 0      low
      1      low
      2    medium
      3    medium
      4    medium
      ...
     103     low
     104     low
     105     low
     106     low
     107    medium
      Name: Temperature, Length: 108, dtype: category
      Categories (4, object): ['low' < 'medium' < 'high' < 'very_high']
```

```
[107]: tidy_data['temp_cat'] = pd.cut(tidy_data['Temperature'], bins=temp_points, labels=temp_labels)
```

```
[108]: tidy_data.head()
```

```
[108]: Parameter      Date      Drug_Name      time  Pressure \
0      15-10-2020  diltiazem hydrochloride  10:30:00      18.0
1      15-10-2020  diltiazem hydrochloride  11:30:00      19.0
2      15-10-2020  diltiazem hydrochloride  12:30:00      20.0
3      15-10-2020  diltiazem hydrochloride   1:30:00      12.0
4      15-10-2020  diltiazem hydrochloride   2:30:00      13.0

Parameter  Temperature  temp_cat
0          20.0        low
1          20.0        low
2          21.0    medium
3          23.0    medium
4          22.0    medium
```

```
[114]: tidy_data['temp_cat'].value_counts()
```

```
[114]: low      45
      medium  30
      high    15
      very_high  5
      Name: temp_cat, dtype: int64
```

```
[110]: # By default, pd.cut() creates intervals of the form (lower_value, upper_value]
```

```
[117]: # Create bins for pressure as well
```

```
[113]: data_melt.head()
```

```
[113]:
```

	Date	Drug_Name	Parameter	time	readings
0	15-10-2020	diltiazem hydrochloride	Temperature	1:30:00	23.0
1	15-10-2020	diltiazem hydrochloride	Pressure	1:30:00	12.0
2	15-10-2020	docetaxel injection	Temperature	1:30:00	NaN
3	15-10-2020	docetaxel injection	Pressure	1:30:00	NaN
4	15-10-2020	ketamine hydrochloride	Temperature	1:30:00	24.0

```
[ ]:
```

0.1.1 Working with missing values:

- NaN -> Not a number. Used for floats/Int.
- None -> Represent missing str values or values in non numeric column

```
[118]: type(None)
```

```
[118]: NoneType
```

```
[120]: type(np.nan)
```

```
[120]: float
```

```
[122]: pd.Series([1, np.nan, 2, None])
```

```
[122]:
```

0	1.0
1	NaN
2	2.0
3	NaN

dtype: float64

```
[125]: # Count the missing values?  
# isna()  
# isnull()
```

```
[128]: data.isna() # count the missing values
```

```
[128]:
```

	Date	Drug_Name	Parameter	1:30:00	2:30:00	3:30:00	4:30:00	5:30:00	\
0	False	False	False	False	False	True	False	False	
1	False	False	False	False	False	True	False	False	
2	False	False	False	True	False	False	True	False	
3	False	False	False	True	False	False	True	False	

4	False	False	False	False	True	True	False	True
5	False	False	False	False	True	True	False	True
6	False	False	False	False	False	False	False	False
7	False	False	False	False	False	False	False	False
8	False	False	False	False	False	True	False	False
9	False	False	False	False	False	True	False	False
10	False	False	False	False	False	False	True	False
11	False	False	False	False	False	False	True	False
12	False	False	False	False	False	False	False	False
13	False	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False	False

	6:30:00	7:30:00	8:30:00	9:30:00	10:30:00	11:30:00	12:30:00
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	True	True	False	False	False	False
3	False	True	True	False	False	False	False
4	False	False	False	False	False	False	False
5	False	False	False	False	False	False	False
6	False	False	False	False	False	True	False
7	False	False	False	False	False	True	False
8	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False
10	False	False	False	True	False	False	False
11	False	False	False	True	False	False	False
12	False	False	True	False	False	False	False
13	False	False	True	False	False	False	False
14	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False

```
[130]: data.isnull() # just an alias for isna()
```

```
[130]:
```

	Date	Drug_Name	Parameter	1:30:00	2:30:00	3:30:00	4:30:00	5:30:00	\
0	False	False	False	False	False	True	False	False	
1	False	False	False	False	False	True	False	False	
2	False	False	False	True	False	False	True	False	
3	False	False	False	True	False	False	True	False	
4	False	False	False	False	True	True	False	True	
5	False	False	False	False	True	True	False	True	
6	False	False	False	False	False	False	False	False	
7	False	False	False	False	False	False	False	False	
8	False	False	False	False	False	True	False	False	

9	False	False	False	False	False	True	False	False
10	False	False	False	False	False	False	True	False
11	False	False	False	False	False	False	True	False
12	False	False	False	False	False	False	False	False
13	False	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False	False

	6:30:00	7:30:00	8:30:00	9:30:00	10:30:00	11:30:00	12:30:00
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	True	True	False	False	False	False
3	False	True	True	False	False	False	False
4	False	False	False	False	False	False	False
5	False	False	False	False	False	False	False
6	False	False	False	False	False	True	False
7	False	False	False	False	False	True	False
8	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False
10	False	False	False	True	False	False	False
11	False	False	False	True	False	False	False
12	False	False	True	False	False	False	False
13	False	False	True	False	False	False	False
14	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False

```
[131]: # Get the count of missing values
```

```
[134]: data.isna().sum()
```

```
[134]: Date          0
Drug_Name         0
Parameter         0
1:30:00           2
2:30:00           2
3:30:00           6
4:30:00           4
5:30:00           2
6:30:00           0
7:30:00           2
8:30:00           4
9:30:00           2
10:30:00          0
```

```
11:30:00    2
12:30:00    0
dtype: int64
```

```
[136]: # row index => count for missing values
data.isna().sum(axis=1)
```

```
[136]: 0      1
      1      1
      2      4
      3      4
      4      3
      5      3
      6      1
      7      1
      8      1
      9      1
     10      2
     11      2
     12      1
     13      1
     14      0
     15      0
     16      0
     17      0
dtype: int64
```

```
[137]: # Calculate total number of missing values in our df
```

```
[ ]:
```

0.1.2 Dropping null values

```
[139]: data.dropna()
```

```
[139]:
```

	Date	Drug_Name	Parameter	1:30:00	2:30:00	\
14	17-10-2020	docetaxel injection	Temperature	12.0	13.0	
15	17-10-2020	docetaxel injection	Pressure	20.0	22.0	
16	17-10-2020	ketamine hydrochloride	Temperature	13.0	14.0	
17	17-10-2020	ketamine hydrochloride	Pressure	8.0	9.0	

	3:30:00	4:30:00	5:30:00	6:30:00	7:30:00	8:30:00	9:30:00	10:30:00	\
14	14.0	15.0	16.0	17	18.0	19.0	20.0	21	
15	22.0	22.0	22.0	23	25.0	26.0	27.0	28	
16	15.0	16.0	17.0	18	19.0	20.0	21.0	22	
17	10.0	11.0	11.0	12	12.0	11.0	12.0	13	

	11:30:00	12:30:00
14	22.0	23
15	29.0	28
16	23.0	24
17	14.0	15

```
[140]: data.dropna(axis=0)
```

```
[140]:
```

	Date	Drug_Name	Parameter	1:30:00	2:30:00	\
14	17-10-2020	docetaxel injection	Temperature	12.0	13.0	
15	17-10-2020	docetaxel injection	Pressure	20.0	22.0	
16	17-10-2020	ketamine hydrochloride	Temperature	13.0	14.0	
17	17-10-2020	ketamine hydrochloride	Pressure	8.0	9.0	

	3:30:00	4:30:00	5:30:00	6:30:00	7:30:00	8:30:00	9:30:00	10:30:00	\
14	14.0	15.0	16.0	17	18.0	19.0	20.0	21	
15	22.0	22.0	22.0	23	25.0	26.0	27.0	28	
16	15.0	16.0	17.0	18	19.0	20.0	21.0	22	
17	10.0	11.0	11.0	12	12.0	11.0	12.0	13	

	11:30:00	12:30:00
14	22.0	23
15	29.0	28
16	23.0	24
17	14.0	15

```
[142]: data.shape
```

```
[142]: (18, 15)
```

```
[141]: data.dropna(axis = 1)
```

```
[141]:
```

	Date	Drug_Name	Parameter	6:30:00	10:30:00	\
0	15-10-2020	diltiazem hydrochloride	Temperature	22	20	
1	15-10-2020	diltiazem hydrochloride	Pressure	14	18	
2	15-10-2020	docetaxel injection	Temperature	18	23	
3	15-10-2020	docetaxel injection	Pressure	23	26	
4	15-10-2020	ketamine hydrochloride	Temperature	26	22	
5	15-10-2020	ketamine hydrochloride	Pressure	9	9	
6	16-10-2020	diltiazem hydrochloride	Temperature	38	40	
7	16-10-2020	diltiazem hydrochloride	Pressure	23	24	
8	16-10-2020	docetaxel injection	Temperature	49	56	
9	16-10-2020	docetaxel injection	Pressure	27	28	
10	16-10-2020	ketamine hydrochloride	Temperature	12	13	
11	16-10-2020	ketamine hydrochloride	Pressure	15	16	
12	17-10-2020	diltiazem hydrochloride	Temperature	16	14	
13	17-10-2020	diltiazem hydrochloride	Pressure	8	11	

14	17-10-2020	docetaxel injection	Temperature	17	21
15	17-10-2020	docetaxel injection	Pressure	23	28
16	17-10-2020	ketamine hydrochloride	Temperature	18	22
17	17-10-2020	ketamine hydrochloride	Pressure	12	13

12:30:00

0	21
1	20
2	25
3	28
4	20
5	11
6	42
7	27
8	58
9	30
10	15
11	18
12	10
13	14
14	23
15	28
16	24
17	15

```
[143]: # data imputation => replace None with 0, mean, median, mode, custome =>
      ↳ upcoming module
```

```
[144]: tidy_data.to_csv('pfizer_tidy.csv', sep=",", index=False)
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```